Notes: Di Maggio and Kacperczyk (JFE, 2017)
The unintended consequences of the zero lower bound policy

Contents

1 Big picture

- **Question.** What happens to money market funds when monetary policy keeps short-term nominal rates at the zero lower bound for a long time? The paper studies whether low rates change money funds' product offerings, exit decisions, fee policy, and broader effects on credit supply and asset management.
- **Setting.** The setting is the U.S. taxable prime money market fund industry from January 2005 to December 2013. Money market funds are useful for this question because their safe short-term assets have returns that closely track the Fed target rate.
- **Core shock.** The paper uses five FOMC events: the December 2008 cut to 0–0.25% and four later forward-guidance announcements indicating that rates would remain near zero for an extended period.
- **Mechanism.** Near-zero safe rates squeeze money funds' gross margins. If funds keep investing safely and still charge normal fees, investor net returns can become unattractive or negative. Funds then face three margins of adjustment: reach for yield, waive fees/subsidize investors, or exit.
- **Main result 1: risk taking.** After zero-rate and forward-guidance shocks, funds increase risk along several dimensions: higher spreads, more risky holdings, and more concentrated portfolios.
- **Main result 2: exit.** The probability of fund closure rises after the policy events. Exit is especially important for affiliated funds whose sponsors face larger reputational spillovers.
- **Main result 3: fee waivers and subsidies.** Funds lower expenses charged to investors even though incurred expenses do not fall much. This creates sponsor subsidies, helping funds deliver non-negative net returns.
- **Cross-sectional result.** Independent sponsors are more likely to respond by taking risk and staying in the market. Affiliated sponsors are more likely to close money funds rather than increase risk.
- **Real and industry effects.** Fund closures reduce credit supplied to issuers that borrowed from the closed funds, temporarily lowering borrower leverage. Fund families that close MMFs are more likely to open bond funds, especially short-term bond funds.
- **Economic takeaway.** Monetary policy at the zero lower bound affects more than aggregate demand and bond yields. It changes the industrial organization, risk profile, pricing, and credit-supply role of an important shadow-banking sector.

2 Incremental contribution of the paper

- **From aggregate ZLB effects to intermediary behavior.** Much of the zero-lower-bound literature studies macro aggregates or asset prices. This paper uses micro-level fund data to show how financial intermediaries adjust their business model under low rates.
- **A new channel of monetary-policy transmission.** The paper identifies a transmission channel through money market funds: low rates compress the profitability of intermediaries that invest in short-term safe assets, inducing risk taking, exits, and fee subsidies.

- **Industrial organization of shadow banking.** The paper does not only ask whether funds take risk. It also studies market structure: fund exits, sponsor decisions, fund-family reallocations, and manager movements.
- **Risk taking versus exit tradeoff.** A central contribution is the equilibrium distinction between two responses to the same shock. Some funds reach for yield, while others shut down. The choice depends on sponsor type and reputational incentives.
- **Direct evidence on sponsor subsidies.** The paper quantifies the difference between incurred and charged expenses, showing that fee waivers are a major hidden adjustment margin in the low-rate environment.
- **Micro-to-real effects.** The paper links fund closures to issuers' outstanding debt and leverage, showing that shocks to MMFs can temporarily reduce credit availability for firms that relied on them.
- **Broader asset-management reallocations.** The paper documents that families closing MMFs open bond funds and reallocate resources to related fixed-income products. This expands the analysis from one fund type to the mutual fund sector as a whole.
- **Identification contribution.** The paper combines event studies around FOMC announcements, cross-sectional sponsor heterogeneity, interest-rate-regime asymmetry, longevity-risk measures, monetary surprises, and security-level holdings evidence.

3 Data and measurement

3.1 Money market fund sample

- The main sample is the universe of U.S. taxable prime money market funds from January 2005 to December 2013.
- The primary data source is iMoneyNet, which provides weekly fund-level information on yields, expenses, maturity, instrument holdings, sponsor identity, and fund characteristics.
- The authors supplement iMoneyNet with CRSP mutual fund data, Compustat, sponsor information from company websites, manager information from several public and proprietary sources, and Federal Reserve data on policy events.
- The unit of analysis is usually the fund portfolio, with share classes aggregated by fund and investor type.
- The sample contains 349 fund portfolios. The average fund size is about \$8.3 billion, average portfolio maturity is about 40 days, and average fund age is about 15.8 years.

Interpretation: The main empirical advantage is high-frequency, fund-level coverage of a sector whose returns are mechanically tied to short-term interest rates. This makes MMFs a clean setting for studying the pressure created by the zero lower bound.

3.2 Policy events and event windows

- The paper focuses on five zero-interest-rate-policy events: the initial December 2008 rate cut and four forward-guidance announcements.

- The forward-guidance events matter because the duration of the low-rate regime determines how long MMF profitability is under stress.
- The main event windows compare one month before the event to either three months or six months after the event.
- The three-month window captures quicker adjustments, such as portfolio risk and fee policy; the six-month window captures slower adjustments, such as fund exit.

Interpretation: The event-study design asks whether fund behavior changes immediately after the Fed tells markets that near-zero rates will continue.

3.3 Sponsor type and reputation measures

- The paper separates funds by sponsor type: affiliated sponsors, such as banks, investment banks, and insurance companies; and independent asset-management companies.
- Affiliated sponsors are expected to have larger reputational spillovers and possibly greater concern about having to support a distressed fund.
- Independent sponsors have stronger incentives to keep MMFs alive by reaching for yield because money-fund business may be more central to them.
- Within independent sponsors, the paper uses *Business risk*, defined as the share of family assets in non-money funds, to proxy for reputation exposure outside the MMF business.

Interpretation: Sponsor heterogeneity is central to the mechanism. The same low-rate shock creates different optimal responses depending on how much the sponsor values reputation and related businesses.

3.4 Main outcome variables

- **Exit.** An indicator equal to one if the fund sponsor closes the fund in a given week.
- **Spread.** Fund return minus the Treasury bill rate; higher values indicate higher return compensation and potentially greater risk.
- **Holdings risk.** The share invested in risky assets, such as bank obligations, net of safer assets such as repos, Treasuries, and agency securities.
- **Maturity risk.** Weighted average portfolio maturity.
- **Concentration.** A Herfindahl index of portfolio holdings across risky asset classes.
- **Charged expenses.** The expense rate charged to investors.
- **Incurred expenses.** The cost rate incurred by the fund.
- **Subsidy.** Incurred expenses minus charged expenses. A larger value means the sponsor is waiving fees or absorbing costs.

Interpretation: The outcomes map directly to the three adjustment margins: risk, exit, and fee subsidies.

3.5 Portfolio holdings, issuer effects, and industry data

- For 2010–2013, the authors use N-MFP filings, which provide monthly security-level MMF holdings, including issuer names, security types, transaction dates, and yields.
- The holdings data allow the authors to test whether funds buy newly added securities with higher yields after policy shocks.
- Issuer-level borrowing data are used to study whether borrowers connected to closed funds experience declines in outstanding debt and leverage.
- CRSP mutual fund data are used to examine whether fund families that close MMFs open funds in other asset classes.
- Manager career data are used to track whether managers of closed funds stay in MMFs, move to other funds, or leave the fund industry.

Interpretation: These additional data let the paper move beyond fund behavior to downstream credit supply and resource reallocation.

4 Models and empirical specifications

4.1 Fund returns and the Fed target rate

The first empirical step establishes that MMF returns are closely linked to the Fed target rate:

$$Fund\ return_{it} = a_0 + a_1 Fed\ rate_t + bX_{it-1} + \varepsilon_{it}. \tag{1}$$

- $Fund\ return_{it}$ is the annualized gross return of fund i in week t.
- $Fed\ rate_t$ is the annualized Fed target rate.
- Controls include fund size, family size, charged expenses, fund age, flows, flow volatility, and an institutional-share indicator.
- Specifications include different combinations of fund, sponsor, and year fixed effects.

Interpretation: This regression verifies the economic premise. If the Fed target rate falls to zero, MMF asset returns and fund gross returns are also pushed toward zero.

4.2 Flow-performance relationship

The paper then tests whether investors respond to fund performance:

$$Fund\ flow_{it} = a_0 + a_1 Fund\ return_{it-1} + bX_{it-1} + \varepsilon_{it}. \tag{2}$$

A second version interacts performance with a low-rate indicator:

$$Fund\ flow_{it} = a_0 + a_1 Fund\ return_{it-1} + a_2(Fund\ return_{it-1} \times Low\ rate_t) + bX_{it-1} + \varepsilon_{it}. \tag{3}$$

- $Low\ rate_t$ equals one when the Fed target rate is at most 1%.
- The interaction tests whether investors become more sensitive to fund returns when rates are near zero.

Interpretation: If performance matters more when rates are low, managers have stronger incentives to alter risk and fees.

4.3 Event-study specification

The main event-study regression is:

$$Fund\ strategy_{it} = a_0 + a_1 Event_t + bX_{i,2006} + \varepsilon_{it}. \tag{4}$$

- $Fund\ strategy_{it}$ is exit, a risk measure, or an expense/subsidy measure.
- $Event_t$ equals one after the policy announcement and zero before it.
- Controls are measured in January 2006 to avoid conditioning on variables that may already have been affected by the monetary shock.
- The paper uses three-month and six-month post-event windows.

Interpretation: The coefficient on $Event_t$ captures the average change in MMF behavior after ZIRP-related policy news.

4.4 Sponsor-type interaction

To test cross-sectional differences, the paper estimates:

$$\begin{aligned} Fund\ strategy_{it} = {} & a_0 + a_1 Event_t + a_2 Independent\ sponsor_i \\ & + a_3 (Event_t \times Independent\ sponsor_i) + bX_{i,2006} + \varepsilon_{it}. \end{aligned} \tag{5}$$

- $Independent\ sponsor_i$ equals one for independent asset-management sponsors.
- The interaction coefficient measures whether independent sponsors respond differently after the shock.

Interpretation: This specification separates the common low-rate shock from differential strategic responses caused by sponsor incentives.

4.5 Reputation within independent sponsors

For independent sponsors, the paper uses:

$$\begin{aligned} Fund\ strategy_{it} = {} & a_0 + a_1 Event_t + a_2 Business\ risk_{i,2006} \\ & + a_3 (Event_t \times Business\ risk_{i,2006}) + bX_{i,2006} + \varepsilon_{it}. \end{aligned} \tag{6}$$

- *Business risk* is the share of family assets in non-money funds.
- The idea is that sponsors with large non-MMF businesses have more to lose from reputational spillovers if their MMFs take risk and fail.

Interpretation: This test holds bailout capacity more fixed and isolates reputation concerns within independent asset managers.

4.6 Interest-rate-regime asymmetry

The paper distinguishes normal-rate and low-rate regimes:

$$Fund\ strategy_{it} = a_0 + a_1 Fed\ rate_t + a_2 Low\ rate_t + a_3(Fed\ rate_t \times Low\ rate_t) + bX_{i,2006} + \varepsilon_{it}. \quad (7)$$

- The core test is whether the response to rate changes is different when the Fed target rate is below 1%.
- A strong effect only near zero supports the ZLB mechanism rather than a generic interest-rate story.

Interpretation: The paper's mechanism requires a margin squeeze near zero, not merely any rate cut.

4.7 Portfolio-holdings and real-effects specifications

For newly added securities, the paper estimates:

$$Mean\ yield_{it} = a_0 + a_1 Event_t + bX_{i,2006} + \varepsilon_{it}. \quad (8)$$

For issuer debt and leverage effects, the paper uses specifications of the form:

$$Outstanding\ debt_{it} = a + \beta Post\ fund\ closure_{it} + \delta_i + \gamma_t + \varepsilon_{it}, \quad (9)$$

$$Leverage_{it} = a + \beta_1(Post_t \times Closure_i) + \beta_2 Closure_i + \beta_3 Post_t + \delta_i + \gamma_t + \varepsilon_{it}. \quad (10)$$

Interpretation: The holdings regressions show active reaching for yield. The issuer regressions ask whether MMF closures reduce borrowers' capital access.

4.8 Industry reallocation specification

To study whether families replace closed MMFs with other funds, the paper estimates:

$$Number\ of\ funds_{it} = a_0 + a_1 After_t + a_2 Treated_i + a_3(After_t \times Treated_i) + bX_{i,2006} + \varepsilon_{it}. \quad (11)$$

- $Treated_i$ equals one for families that close an MMF.

- $After_t$ equals one after the closure window.
- The dependent variable is the number of funds in money, bond, equity, balanced, or bond-maturity categories.

Interpretation: This specification asks whether the low-rate shock reshapes the product menu of fund families.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies changes in MMF behavior around monetary policy announcements that altered the expected duration of near-zero rates.
- The high-frequency weekly data help narrow the event window and reduce contamination from slower-moving macro trends.
- Cross-sectional tests compare funds exposed to the same macro shock but with different sponsor incentives.
- The security-level evidence shows that funds actively bought higher-yielding securities after the policy events, rather than the results being only mechanical portfolio drift.

Interpretation: The evidence is strongest as a micro-level event-study analysis of intermediary responses to ZLB-related news.

5.2 Why the zero bound is special

- In a normal-rate environment, lower rates do not necessarily force MMFs into negative net-return territory.
- Near zero, however, safe assets cannot easily generate enough gross return to cover normal expenses.
- The paper tests this by comparing responses above and below a 1% Fed target-rate threshold.
- Results are much stronger in the low-rate regime, supporting the view that the zero lower bound creates a discontinuous business-model problem for MMFs.

Interpretation: The key empirical claim is not simply that rate cuts matter. It is that rate cuts close to zero change the constraints faced by MMFs.

5.3 Selection and survival concerns

- Average risk can rise because safer funds exit, leaving riskier funds in the sample.
- Average risk can also rise because surviving funds actively change their portfolios.
- The authors address this by repeating risk tests on the sample of funds that survive across the event windows.

- The results remain similar, implying that both selection and active strategic adjustment contribute to the increased average risk.

Interpretation: The fund-risk result is not only a composition effect. Surviving funds also reach for yield.

5.4 Alternative macro explanations

- FOMC announcements may respond to macroeconomic conditions, so policy events are not fully random.
- The paper mitigates this concern with narrow windows, fixed effects, controls measured before the crisis, monetary policy surprise measures, and placebo dates.
- QE events are not the focus because QE primarily targeted long-maturity assets, while MMFs are restricted to short-term high-quality assets.

Interpretation: The identification is not perfect randomization, but the paper uses multiple tests to separate ZLB pressure from broader macro trends.

5.5 Main limitations

- The design is observational and event-study based; it cannot completely rule out all concurrent shocks.
- Sponsor type may capture more than reputation, including access to support, business model, distribution, and client base.
- Issuer-level real effects are measured on a matched subset, so external validity should be interpreted carefully.
- The welfare implications are ambiguous: fee waivers help investors, but reaching for yield and exits can increase financial fragility and reduce short-term credit supply.

Interpretation: The paper provides strong evidence of strategic adjustment, but a full welfare analysis remains outside the paper.

6 Tables and figures

6.1 Table 1: Zero Interest Rate Policy Events (ZIRP)

Read:

- Table 1 lists the five monetary-policy events used in the event-study design.
- The first event, December 16, 2008, is the reduction of the Fed target rate to 0–0.25%.
- The next four events are forward-guidance announcements indicating that near-zero rates would continue for an extended period or until future years.

- The table clarifies that the shock is not only the level of rates, but also the expected duration of the low-rate regime.

Economic message: The table defines the empirical shock. Once the Fed commits to keeping rates low for longer, money funds face a longer profitability squeeze.

Table 1
Zero Interest Rate Policy Events (ZIRP).
We report the dates of FOMC meetings in which the Fed decided to change the Fed target rate or provided policy guidance about the prevailing zero interest rate policy.

Date	Event
December 16, 2008	Fed target rate reduced to 0–0.25%
March 18, 2009	Zero rates for "an extended period of time"
August 9, 2011	Zero rates at least until 2013
January 25, 2012	Zero rates at least until 2014
September 13, 2012	Zero rates at least until 2015

Table 1: Zero Interest Rate Policy Events.

6.2 Table 2: Fund gross returns and Fed target rate

Read:

- Table 2 shows that MMF gross returns are highly sensitive to the Fed target rate.
- In specifications with fund or sponsor fixed effects, the coefficient on the Fed rate is large and strongly significant.
- The estimates imply that when the Fed target rate falls, MMF gross returns fall almost mechanically.
- The result validates the institutional premise that MMFs are directly exposed to short-term monetary policy.

Economic message: This is the first building block of the mechanism. If the Fed target rate is near zero, the traditional safe-asset business model of prime MMFs is put under stress.

Table 2
Fund gross returns and Fed target rate.

The sample is all U.S. prime money market funds over the period January 2005–December 2013. The dependent variable is *Fund gross return* computed as the annualized return. *Fed rate* is the annualized Fed target rate. Control variables include the natural logarithm of fund assets, the natural logarithm of family assets, the expense ratio (charged), fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow growth, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level. Column 1 includes year-fixed effects, column 2 includes fund-fixed effects, column 3 includes sponsor-fixed effects, and column 4 includes year-fixed and sponsor-fixed effects. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)
Fed rate	93.025***	94.370***	62.291***	62.086***
	(1.012)	(0.963)	(5.087)	(5.083)
Controls	Yes	Yes	Yes	Yes
Year-fixed effects	No	No	Yes	Yes
Fund-fixed effects	Yes	No	No	No
Sponsor-fixed effects	No	Yes	No	Yes
Observations	98,496	98,496	98,496	98,496

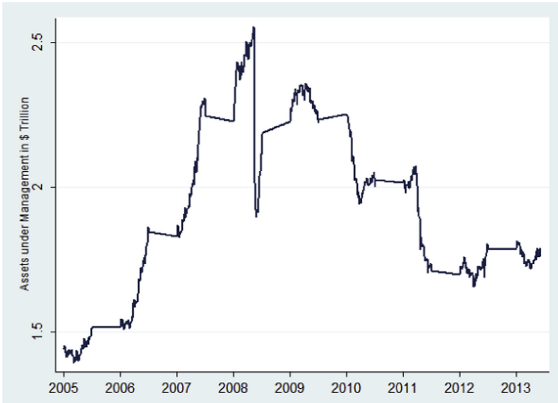
Table 2: Fund gross returns and Fed target rate.

6.3 Figure 1 and Table 3: Assets under management and flow-performance sensitivity

Read:

- Figure 1 plots total assets under management of prime money funds from 2005 to 2014.
- Assets rise before the crisis, then decline sharply during the low-rate period.
- Table 3 shows that fund flows respond positively to fund returns.
- The interaction of fund return with the low-rate indicator is positive, meaning investors are more return-sensitive when rates are near zero.
- The combination of the figure and table explains why funds have incentives to protect investor returns by taking risk or waiving fees.

Economic message: Low returns matter because investors move money away from weak-performing funds. The zero-rate environment increases the payoff to any fund strategy that preserves returns.



(a) Figure 1: Prime MMF assets under management.

Table 3
The flow-performance relationship.

The sample is all U.S. prime money funds over the period January 2005–December 2013. The dependent variable is *Fund flow*, computed as the percentage change in total net assets from time t to time $t+1$, adjusted for market appreciation. *Fed rate* is the annualized Fed target rate. *Fund return* is the annualized fund return. Control variables include the natural logarithm of fund assets, the natural logarithm of family assets, expense ratio (charged), fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level and include sponsor-fixed effects. *Low rate* restricts the sample to the period of low interest rates (Fed target rate between zero and 1%). Columns 2 and 4 additionally include week-fixed effects. Standard errors are clustered at the fund sponsor level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)
	Flow		Flow	
Fund return	0.001***	0.009***	0.002***	0.006***
	(0.000)	(0.001)	(0.000)	(0.002)
Fund return×Low rate			0.002**	0.006***
			(0.001)	(0.002)
Controls	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes
Week-fixed effects	No	Yes	No	Yes
Observations	98,948	98,948	98,948	98,948

(b) Table 3: Flow-performance relationship.

Figure 1 and Table 3: Investor flows and money-fund returns.

6.4 Table 4: Fund strategies and ZIRP shocks

Read:

- Panel A shows that the number of funds falls after ZIRP-related announcements, and exit probability rises.
- Panel B shows risk taking increases: spreads, holdings risk, and concentration rise after the events.
- Maturity falls in the event window, likely reflecting regulatory and liquidity constraints rather than lower risk overall.
- Panel C shows charged expenses decline, especially over the six-month horizon.
- Subsidies rise in the six-month window because funds reduce charged fees more than incurred costs.

Economic message: Table 4 is the main time-series result. The low-rate shock pushes funds toward three strategic responses: exit, reach for yield, and waive fees.

Table 4
Fund strategies and ZIRP shocks.

The sample is all U.S. prime MMFs. The estimation window includes one month before and three months or six months after the event dates defined in Table 1. *Event* is an indicator variable equal to one for the period after the event date and zero for the period before the event date. In Panel A, the dependent variables are *Number of funds*, defined as the number of funds in a given period, and *Exit*, defined as an indicator variable equal to one if the fund exits the fund industry in week t. In Panel B, the dependent variables are: the weekly annualized spread (*Spread*), the fraction of assets held in risky assets, net of the riskless assets (*Holdings risk*), average portfolio maturity (*Maturity risk*), and portfolio concentration, defined as a Herfindahl Index of asset classes (*Concentration*). In Panel C, the dependent variables are *Charged expenses*, defined as percentage expense rate charged by a fund, *Incurred expenses*, defined as percentage expense rate incurred by a fund, *Subsidy*, defined as the difference between incurred and charged expenses. Control variables include the annualized fund return, the natural logarithm of fund assets, the natural logarithm of family assets, expense ratio (charged), fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level and include year-fixed and sponsor-fixed effects. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance.

Panel A: Fund exit

	(1)	(2)	(3)	(4)
	# Funds	Exit	# Funds	Exit
	3-Months ahead		6-Months ahead	
Event	−5.400***	0.002**	−9.000***	0.002**
	(0.543)	(0.001)	(1.152)	(0.001)
Controls	No	Yes	No	Yes
Year-fixed effects	No	Yes	No	Yes
Sponsor-fixed effects	No	Yes	No	Yes
Observations	10	18,568	10	25,914

Panel B: Fund risk

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Spread	Holding risk	Maturity	Concentration	Spread	Holding risk	Maturity	Concentration
	3-Months ahead				6-Months ahead			
Event	34.329***	0.954***	−1.216**	0.005***	21.341**	0.879***	−1.347***	0.005***
	(12.113)	(0.323)	(0.503)	(0.002)	(10.471)	(0.271)	(0.386)	(0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,830	16,882	16,879	16,882	23,462	23,527	23,524	23,527

Panel C: Fund expenses

	(1)	(2)	(3)	(4)	(5)	(6)
	Charged	Incurred	Subsidy	Charged	Incurred	Subsidy
	3-Months ahead			6-Months ahead		
Event	−0.001	−0.002**	−0.002	−0.014**	−0.003***	0.010**
	(0.002)	(0.001)	(0.002)	(0.006)	(0.001)	(0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,568	18,506	18,506	25,914	25,829	25,829

Table 4: Fund strategies and ZIRP shocks.

6.5 Table 5: Conditioning on sponsor type

Read:

- Table 5 interacts the event indicator with an independent-sponsor indicator.
- Independent funds are less likely to exit after the shocks, especially over the six-month horizon.
- Independent sponsors take more risk after the shocks across several risk measures.
- Expense-policy responses do not differ much by sponsor type.

- The results imply a tradeoff: affiliated sponsors tend to close funds, while independent sponsors tend to stay and reach for yield.

Economic message: Sponsor incentives shape how the same monetary shock is transmitted. Reputation-sensitive affiliated sponsors avoid risk by exiting; independent sponsors preserve business by increasing risk.

Table 5
Fund strategies and ZIRP shocks: Conditioning on sponsor type.
The design follows Table 5. *Independent sponsor* is an indicator variable equal to one if the fund sponsor is an independent asset management company, and zero otherwise.

Panel A: Fund exit

	(1)	(2)	(3)	(4)
	# Funds	Exit	# Funds	Exit
	3-Months ahead		6-Months ahead	
Event	−3.280***	0.002***	−2.862***	0.002***
	(0.803)	(0.001)	(0.767)	(0.001)
Independent sponsor×Event	0.077	−0.002	0.178**	−0.003**
	(0.067)	(0.002)	(0.081)	(0.001)
Controls	Yes	Yes	Yes	Yes
Year-fixed effects	No	Yes	No	Yes
Sponsor-fixed effects	No	Yes	No	Yes
Observations	20	18,568	20	25,914

Panel B: Fund risk

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Spread	Holding risk	Maturity	Concentration	Spread	Holding risk	Maturity	Concentration
	3-Months ahead				6-Months ahead			
Independent sponsor	−3.837	6.642**	4.768***	0	−0.776	5.681	5.107***	−0.011
	(2.878)	(3.309)	(1.366)	(0.020)	(2.940)	(3.492)	(1.484)	(0.021)
Independent	9.215***	1.821***	−1.010**	0.008**	4.646***	3.405**	−1.897***	0.026***
Sponsor×Event	(1.239)	(0.634)	(0.460)	(0.004)	(1.668)	(1.533)	(0.709)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Y/M-F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,830	16,882	16,879	16,882	23,462	23,527	23,524	23,527

Panel C: Fund expenses

	(1)	(2)	(3)	(4)	(5)	(6)
	Charged	Incurred	Subsidy	Charged	Incurred	Subsidy
		3-Months ahead			6-Months ahead	
Independent sponsor	−0.003	0.030	0.032	−0.006	0.029	0.034
	(0.020)	(0.039)	(0.027)	(0.022)	(0.039)	(0.027)
Independent sponsor×Event	0.005	0.008	0.003	0.011	0.011	0
	(0.006)	(0.005)	(0.007)	(0.012)	(0.010)	(0.013)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Week-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,568	18,506	18,506	25,914	25,829	25,829

Table 5: Fund strategies and ZIRP shocks by sponsor type.

6.6 Table 6: Variation in reputation within independent sponsors

Read:

- Table 6 restricts attention to independent sponsors and uses business risk as a measure of reputation exposure outside MMFs.
- Sponsors with greater non-money-fund business take less additional risk after the event.
- The interaction of event and business risk is negative for spread and holdings risk, especially over the six-month horizon.
- Sponsors with larger reputation exposure provide more subsidies.
- Exit effects are not as strong in this within-independent-sponsor test.

Economic message: Even among independent sponsors, reputation matters. Sponsors with more valuable non-MMF franchises choose subsidies over risk taking.

Table 6
Variation in reputation within independent fund sponsors.
The design follows Table 5. *Business risk* is the ratio of assets under management held by a fund family in non-money funds and total assets under management. Panel A presents the results for the 3-month announcement horizon and Panel B for the 6-month horizon.

Panel A: 3-Month horizon

	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Event	0.001	25.207**	4.368*	−0.120	0.018*	−0.050***
	(0.002)	(12.275)	(2.439)	(0.936)	(0.010)	(0.015)
Event×	−0.001	−11.003*	−3.661*	−1.774*	−0.016	0.062***
Business risk	(0.003)	(6.241)	(2.349)	(1.002)	(0.012)	(0.019)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7635	7030	7036	7036	7036	7619

Panel B: 6-Month horizon

	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Event	−0.001	22.596**	7.848***	2.395**	0.036***	−0.052***
	(0.004)	(10.835)	(2.016)	(0.958)	(0.008)	(0.011)
Event×	−0.001	−11.003**	−8.226***	−4.705***	−0.039***	0.079***
Business risk	(0.005)	(4.826)	(2.367)	(1.113)	(0.009)	(0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,659	9789	9802	9802	9802	10,633

Table 6: Reputation variation within independent fund sponsors.

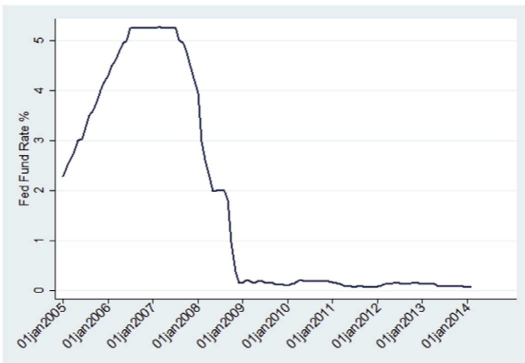
6.7 Figure 2 and Table 7: Interest-rate regimes

Read:

- Figure 2 shows the Fed target rate rising before the crisis, falling sharply in 2007–2008, and staying near zero afterward.
- Table 7 tests whether fund responses differ when the Fed target rate is above or below 1%.

- The strongest effects appear in the low-rate regime, not in normal-rate periods.
- When rates move from 1% toward zero, exit probability, risk taking, and subsidies rise.
- The asymmetric response supports the paper's claim that the zero lower bound creates a special form of pressure.

Economic message: The key problem is not a generic rate decline. It is the compression of net returns when safe short-term rates approach zero.



(a) Figure 2: Fed target rate, 2005–2014.

Table 7
Comparisons across interest rate regimes.

The sample is all U.S. prime money market funds over the period January 2005–December 2013. The definitions of all dependent and control variables follow those in Table 5. All regressions are at the weekly level and include year/month-fixed and sponsor-fixed effects. *Low rate* is an indicator variable equal to one if Fed target rate is below 1%, and zero otherwise. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A: Fund exit

	(1)	(2)	(3)	(4)
	# Funds		Exit	
			RF>1	RF<=1
Fed rate	−1.555***	−2.348***	0.003	−0.072***
	(0.475)	(0.258)	(0.002)	(0.016)
Low rate		−132.783***		
		(12.459)		
Fed rate×Low rate		16.861***		
		(3.171)		
Controls	Yes	Yes	Yes	Yes
Year/Month-F. E.	No	No	Yes	Yes
Sponsor F. E.	No	No	Yes	Yes
Observations	442	442	50,334	48,458

Panel B: Fund risk

	(1)	(2)	(3)	(4)
	Spread	Holdings risk	Maturity risk	Concentration
Fed rate	2.713	−0.048	−0.893	0.001
	(26.732)	(0.28)	(0.649)	(0.001)
Fed rate×Low rate	−56.611***	−5.941***	−1.673*	−0.028***
	(13.628)	(0.723)	(1.032)	(0.006)
Controls	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes
Year/Month-fixed effects	Yes	Yes	Yes	Yes
Observations	94,521	95,264	95,253	94,264

Panel C: Fund expenses

	(1)	(2)	(3)	(4)	(5)	(6)
	Charged expenses		Incurred expenses		Subsidy	
Fed rate	−0.006***	−0.005***	−0.002	0.004***	0.004***	0.008***
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
Fed rate×Low rate		0.080***		−0.001		−0.079***
		(0.027)		(0.011)		(0.022)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	98,484	98,484	98,484	98,484	98,484	98,484

(b) Table 7: Comparisons across interest-rate regimes.

Figure 2 and Table 7: Why the low-rate regime is different.

6.8 Table 8: The effect of longevity risk

Read:

- Table 8 uses a market-implied measure of how long rates are expected to remain below 1%.
- Longer expected duration of low rates predicts higher spreads, higher holdings risk, higher concentration, and larger subsidies.

- The exit coefficient is positive but not statistically strong.
- The evidence shows that expectations about the duration of ZIRP matter, not only the current rate level.

Economic message: The longer the low-rate regime is expected to last, the less viable the traditional MMF model becomes, and the stronger the incentives to reach for yield or subsidize investors.

Table 8
The effect of longevity risk.
The sample is all U.S. prime MMFs over the period January 2005–December 2013. The definitions of all dependent variables follow those in Table 5. *Longevity* is defined as the natural logarithm of 1 + the number of days that would take for the interest rate (30-day) to exceed 1%. The values are interpolated off the future yield curve. All regressions are at the weekly level and include year/month-fixed and sponsor-fixed effects. We focus on the case in which the Fed target rate is below 1%. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Longevity	0.0012	29.937***	0.0090***	−0.193	0.00300***	0.030***
	(0.0008)	(7.046)	(0.0017)	(0.193)	(0.00120)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	48,458	45,322	46,286	46,054	46,065	48,298

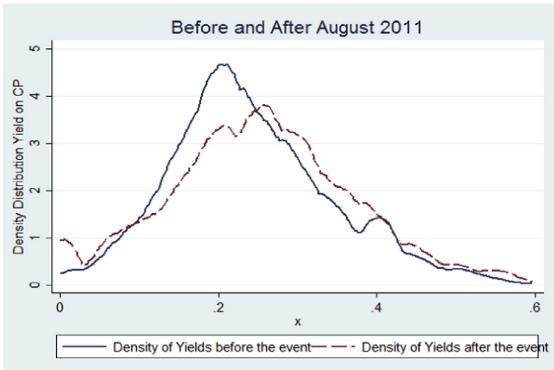
Table 8: The effect of longevity risk.

6.9 Figure 3: Yield distributions around ZIRP shocks

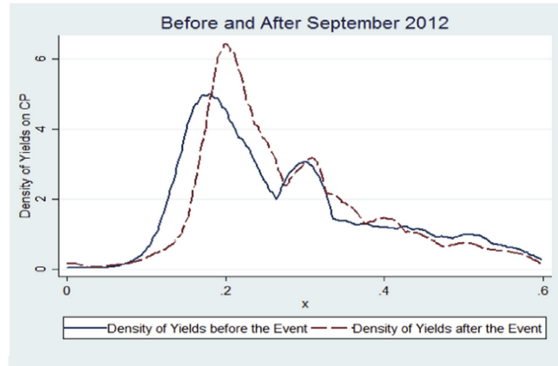
Read:

- Figure 3 compares distributions of yields on commercial paper holdings before and after three ZIRP shocks.
- In each panel, the after-event distribution shifts toward higher yields.
- This supports the interpretation that funds actively buy higher-yielding securities after policy announcements.
- The figure is important because it uses security-level holdings rather than only fund-level averages.

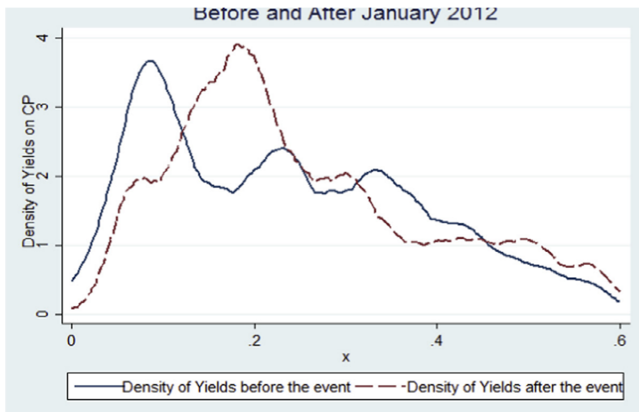
Economic message: Reaching for yield is visible directly in portfolio composition. Funds do not merely experience higher yields passively; they rebalance toward securities with higher yields after the shocks.



(a) Before and after August 2011.



(b) Before and after September 2012.



(c) Before and after January 2012.

Figure 3: Yield distributions of commercial paper around ZIRP shocks.

6.10 Table 9: Evidence from portfolio holdings and real effects

Read:

- Panel A shows that newly added securities have higher yields after the three ZIRP shocks.
- Placebo dates do not show the same pattern; if anything, placebo coefficients are negative.
- Panel B shows the number of borrowers per fund declines after events, indicating changes in issuer composition.
- Panel C shows outstanding debt falls after fund closure for issuers connected to closed funds.
- Panel D links closures to borrower leverage: leverage declines over the six-month window, but the one-year effect is not statistically significant.

Economic message: Table 9 connects the fund response to real credit supply. Fund closures and portfolio changes do not stay inside the mutual fund industry; they temporarily affect issuers that relied on MMFs for short-term funding.

Table 9
Evidence from portfolio holdings.

The sample is all U.S. prime MMFs. The dependent variable is the mean of the yields of the new securities added to the fund portfolio. The estimation window includes one month before and three months after the last three event dates defined in Table 1. *Event* is an indicator variable equal to one for the period after the event date and zero for the period before the event date. Columns 4 and 5 repeat the analysis by considering a randomly drawn date that does not coincide with any of the ZIRP shocks dates. All regressions are at the monthly level and include year/month-fixed and fund-fixed effects. Standard errors are clustered at the monthly level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A

	(1)	(2)	(3)	(4)	(5)
	Mean yield				
				Placebos	
	Event: August 2011	*Event: January 2012*	*Event: September 2012*	*March 2011*	*January 2013*
Event	0.0388***	0.0439***	0.0420***	−0.0302***	−0.0107***
	(0.000701)	(0.000971)	(0.000371)	(0.000438)	(0.000287)
Controls	Yes	Yes	Yes	Yes	Yes
Year/month-fixed effects	Yes	Yes	Yes	Yes	Yes
Fund-fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	49,242	39,690	59,716	31,988	24,129

Table 9, Panel A: Mean yields of newly added securities.

Panel B

Number of borrowers per fund	(1)	(2)
Event	−0.423**	−0.378**
	(0.209)	(0.182)
Fund-fixed effects	Yes	Yes
Time trend	No	Yes
Observations	1,288	1,288
R-squared	0.902	0.902

(a) Panel B: Borrowers per fund.

Panel C

	(1)	(2)
	Outstanding debt	
Post fund closure	−0.419***	−0.153**
	(0.0686)	(0.0665)
Issuer-fixed effects	Yes	Yes
Month-fixed effects	No	Yes
Observations	52,682	52,682
R-squared	0.492	0.524

(b) Panel C: Outstanding debt.

Table 9, Panels B and C: Borrower composition and debt effects.

Panel D

	(1) *Leverage 6-month window*	(2) *Leverage 6-month window*	(3) *Avg. leverage 6-month window*	(4) *Leverage 1-year window*
Post × Closure	−0.006**	−0.006*	−0.020*	−0.004
	(0.003)	(0.003)	(0.011)	(0.008)
Post	0.001	−0.000	0.007	0.002
	(0.002)	(0.001)	(0.004)	(0.007)
Closure	0.004	0.013**	0.016*	0.003
	(0.012)	(0.005)	(0.008)	(0.004)
Time-fixed effects	Yes	Yes	Yes	Yes
Issuer-fixed effects	No	Yes	Yes	Yes
Observations	895	895	448	1,784
R-squared	0.005	0.918	0.929	0.909

Table 9, Panel D: Leverage effects.

6.11 Figure 4 and Table 10: Industry response

Read:

- Figure 4 compares AUM in money funds and bond funds from 2005 to 2014.
- Money fund assets decline during the low-rate period, while bond fund assets grow.
- Table 10 formally tests whether families that close MMFs open funds in other asset classes.
- Families that close MMFs are more likely to open bond funds in the low-rate regime.
- The strongest bond-fund response is in short-term bond funds, with some weaker evidence for long-term bond funds.
- The paper does not find similar strong evidence for equity or balanced fund creation.

Economic message: The zero lower bound reallocates resources within asset management. Fund families shift away from stressed money funds toward nearby fixed-income substitutes.

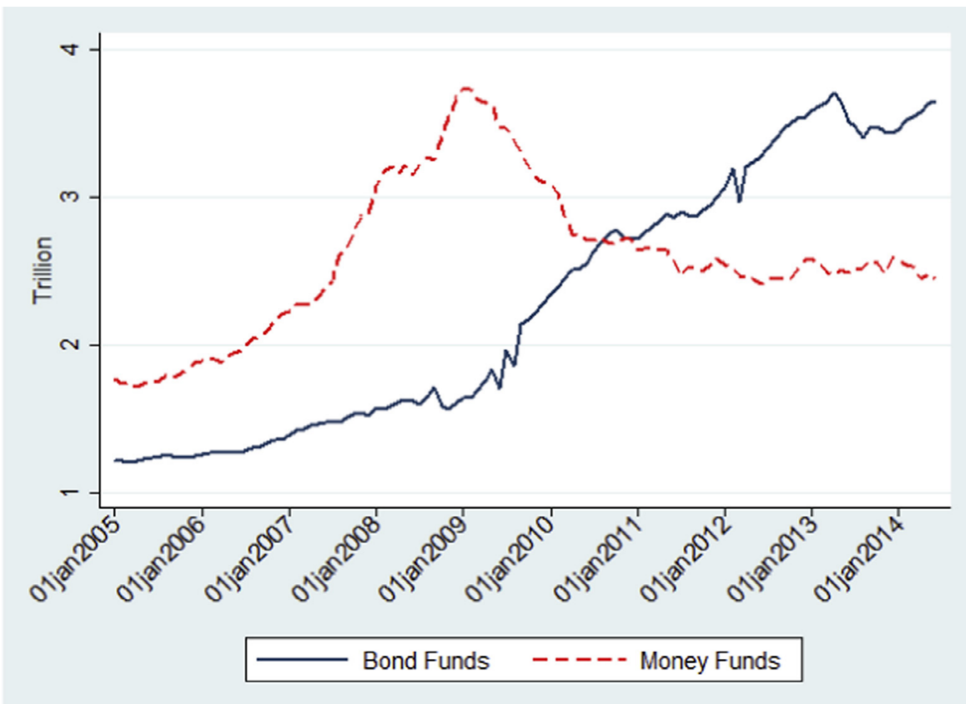


Figure 4: Money funds versus bond funds AUM.

Table 10
Industry response.

The sample includes all mutual funds families in U.S. over the period January 2005–December 2013. The dependent variable is the number of funds (at share class level) in a given period for a given type of fund within a fund family. Panel A groups funds into money, bond, equity, and balanced. Panel B divides bond funds into short-term, medium-term, and long-term. LR denotes low interest rate regime and HR denotes high interest rate regime. *Treated* is an indicator variable equal to one if the fund family closed one of its MMFs and zero for all other funds. *After* is an indicator variable equal to one six months after the closure of the MMF and zero six months before the fund closure. Control variables are family return and natural logarithm of family size. All regressions include fund family-fixed effects. Standard errors are clustered at the year-month level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A: Asset allocations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Money funds		Bond funds		Equity funds		Balanced funds	
	LR	HR	LR	HR	LR	HR	LR	HR
After	−0.113	0.000	0.070	−0.068	0.420	−0.275	0.174	0.296
	(0.068)	(0.500)	(0.206)	(0.198)	(0.654)	(0.546)	(0.158)	(0.185)
After×Treated	−3.313**	−0.688	7.098**	−1.032	16.932	0.818	3.944*	0.643
	(1.528)	(1.657)	(3.620)	(1.987)	(13.802)	(3.455)	(2.255)	(1.181)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Family-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,951	2,834	2,951	2,834	2,951	2,834	2,951	2,834

Panel B: Bond maturity

	(1)	(2)	(3)	(4)	(5)	(6)
	Short-term bonds		Medium-term bonds		Long-term bonds	
	LR	HR	LR	HR	LR	HR
After	0.105	0.090**	0.765*	−4.410	−0.799**	4.251
	(0.067)	(0.041)	(0.391)	(3.095)	(0.316)	(2.969)
After×Treated	0.757**	0.397	1.604	−3.678	4.738*	2.249
	(0.336)	(0.479)	(1.103)	(9.684)	(2.669)	(9.883)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Family-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,951	2,834	2,951	2,834	2,951	2,834

Table 10: Industry response.

6.12 Table 11: The effect of MMF closure on managers' rotation

Read:

- Table 11 tracks managers after their MMF closes.
- About 32.4% stay in the same family and run another MMF.
- About 16.0% stay in the same family but move to a non-MMF.
- About 18.6% move to another family and run an MMF, while 11.7% move to another family and run a non-MMF.
- About 21.3% move to another finance career.
- The table suggests that many managers do not simply remain in the same MMF role after closures.

Economic message: The growth of bond funds after MMF closures is more consistent with product-demand and family-level resource reallocation than with simply finding a new seat for each displaced MMF manager.

Table 11
The effect of MMF closure on managers' rotation.

The sample is conditional on the fund closing its operations. We track managers' subsequent career outcomes. Possible career outcomes are: (1) staying in the same family and running a MMF, (2) staying in the same family and running a non-money market fund, (3) going to a different family and running a MMF, (4) going to a different family and running a non-money market fund, (5) leaving the fund industry (moving to private business, government institutions, setting up own company, etc.). We condition on a single manager departure in a given date (a particular multiple fund departure is called as one case).

Career outcome	MMF in the same family	Non-MMF in the same family	MMF in another family	Non-MMF in another family	Other finance career
# of Cases	61	30	35	22	40
% of Cases	32.4%	16.0%	18.6%	11.7%	21.3%

Table 11: Managers' rotation after MMF closure.

7 Short summary

- The paper studies how the zero lower bound and forward guidance affected U.S. prime money market funds.
- MMFs are an ideal setting because their assets are short-term, safe, and closely tied to the Fed target rate.
- When rates approach zero, MMFs face a margin squeeze: safe assets earn too little to cover normal fees while keeping investor returns attractive.
- The paper identifies three main adjustment margins: reaching for yield, exiting the market, and waiving fees through subsidies.
- After ZIRP-related events, funds become more likely to exit and surviving funds increase risk.
- Charged expenses fall while incurred expenses do not fall as much, implying economically meaningful sponsor subsidies.
- Independent sponsors are more likely to stay and increase risk; affiliated sponsors are more likely to exit.
- Within independent sponsors, those with more non-MMF business are less willing to take risk and more willing to subsidize.
- The response is concentrated in the low-rate regime, supporting the zero-bound mechanism rather than a generic rate-change story.
- Security-level holdings show that newly added securities have higher yields after policy shocks.
- Fund closures reduce issuers' outstanding debt and temporarily reduce leverage among affected borrowers.
- Fund families closing MMFs are more likely to open bond funds, especially short-term bond funds.
- The paper's main contribution is to show that unconventional monetary policy can reshape the risk taking, survival, pricing, and product design of shadow-banking intermediaries.

8 Conclusion

8.1 Core conclusion

Di Maggio and Kacperczyk show that the zero lower bound policy had important unintended consequences for the U.S. money fund industry. When short-term rates were kept near zero, prime MMFs faced a direct profitability shock. They responded by increasing portfolio risk, reducing fees charged to investors, receiving subsidies from sponsors, and in many cases exiting the market.

8.2 Economic intuition

The economic intuition is straightforward. Money funds traditionally offer safety, liquidity, and modest yield by holding short-term safe assets. When the Fed target rate falls to zero, the yields on those assets collapse. If a fund still charges normal expenses, investor net returns become unattractive. To prevent outflows, the fund can either find higher-yielding assets, waive fees, or shut down. These choices are shaped by sponsor incentives. Independent sponsors have stronger reasons to keep the fund business alive, while affiliated sponsors may prefer exit to the reputational cost of taking risk.

8.3 Incremental contribution revisited

The paper advances the literature by showing a micro-level intermediary channel of unconventional monetary policy. It does not merely document that low rates are associated with more risk taking. It shows how fund sponsors choose among risk taking, fee subsidies, and market exit; how these choices differ by sponsor type; how portfolio holdings change after policy events; and how closures affect borrowers and product offerings across the mutual fund industry.

8.4 Implications for monetary policy and financial stability

The findings imply that prolonged low rates can stimulate the economy while also creating pressure in sectors whose business models depend on positive short-term rates. In MMFs, this pressure can increase exposure to credit, liquidity, and concentration risk. It can also reduce short-term funding to issuers when funds exit. For regulators, the paper highlights the importance of monitoring not only banks, but also nonbank intermediaries whose incentives change under unconventional policy.

8.5 Implications for asset-management strategy

For fund families, the paper suggests that low-rate environments force strategic product decisions. Families can subsidize existing MMFs, shift risk within MMFs, close funds, or redirect resources toward bond funds. The evidence that bond funds expand after MMF closures suggests that fund families treat fixed-income products as nearby substitutes for money fund investors and management resources.

8.6 Limitations

The analysis is observational and uses event-study variation rather than randomized assignment. Monetary policy announcements may be correlated with broader macroeconomic conditions, although the paper uses several robustness tests to address this concern. Sponsor type may proxy for multiple organizational differences beyond reputation. The issuer-level real effects are based on matched borrowers and should be interpreted as evidence of temporary credit-supply effects rather than a complete welfare assessment.

8.7 Takeaway

The enduring takeaway is that the zero lower bound changes the incentives and organization of financial intermediaries. For money market funds, near-zero rates do not simply lower returns; they alter risk taking, fees, survival, credit supply, and the allocation of resources across the asset-management industry. The paper turns the broad concern about "reaching for yield" into a detailed empirical account of how a specific shadow-banking sector adjusts when safe short-term rates disappear.